PETER MUSILA

Nairobi, Kenya | +254 719443774 | musiladev@gmail.com linkedin.com/in/peter-musila | github.com/peter-kiilu

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Professional Summary

Innovative MERN stack Junior Software Engineer and Data Science specialist focused on building intelligent systems and scalable digital products. Proven ability to architect full-stack applications across web and mobile development, implement AI-powered solutions (RAG, LLMs, agentic workflows), and transform raw data into actionable insights. Experienced in leading technical initiatives, collaborating with cross-functional teams, and delivering measurable impact.
Passionate about leveraging technology to solve complex, real-world problems.

Technical Skills

- **Full-Stack Development:** React (TypeScript), Python (Flask), Node.js, JavaScript, PostgreSQL, RESTful API Design, System Architecture, Software Design Patterns, MongoDB, Supabase
- **AI & Data Science:** Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (Chroma), Machine Learning, Exploratory Data Analysis (EDA), Data Preprocessing & Cleaning, Prompt Engineering, AI Agents & Automations
- **Cloud & DevOps:** Docker, Kubernetes, CI/CD Pipelines, Cloud-Native Development
- **Core Competencies:** AI Ethics & Literacy, Distributed Architecture, System Design, Critical Thinking, Cross-functional Team Leadership, Project Management, Technical Documentation

Certifications

- Building AI Agents & Automations – Build Fast With AI (Feb 2026)
- AI Literacy – IBM Student Ambassador (Mar 2026)

- Analyzing Data with PromptBI – PromptBI (Nov 2025)
- Exploratory Data Analysis – Martial Jenga Labs (Jan 2025)

Professional Experience

AI Software Developer | Tech Innovators Network (THiNK)
Jan 2026 – Mar 2026 | Embu, Kenya

- Built and deployed a RAG chatbot using Python, LangChain, ChromaDB, PostgreSQL; indexed 290+ docs, achieved 95% accuracy, <2s response time.
- Automated content ingestion and semantic retrieval, saving 45 hours/week; built React frontend with secure auth for 20+ active users.
- Participated in code review sessions to maintain code quality and ensure adherence to best practices.
- Created technical documentation for the chatbot architecture, API endpoints, and deployment, enabling seamless handover and future maintenance.

Founder & Leader | Responsible Computing Young Innovators Club
Jan 2024 – Present | Embu, Kenya

- Led ethical AI initiatives; mentored students in building responsible tech solutions.
- Boosted engagement **60%** via training sessions and innovation challenges.
- **Translated complex AI concepts for non-technical audiences** during workshops and club presentations, helping students from diverse academic backgrounds grasp advanced topics and apply them to real-world problems.

Product Developer (Intern) | MarketForce
Jul 2024 – Feb 2025 | Nairobi, Kenya

- Contributed to scalable product design across the **software development life cycle (SDLC)** , from requirements gathering to deployment.
- Participated in **agile development** processes, including sprint planning, stand-ups, and retrospectives, to enhance project delivery.

- Supported distributed architectures and **cloud-native development** workflows (Docker, Kubernetes, CI/CD).
- Assisted in **software testing**, debugging, and performance optimization for productionready systems.
- Utilized **software design patterns** to improve code structure and maintainability.

ICT Support Officer (Intern) | University of Embu
May 2025 – Aug 2025 | Embu, Kenya

- Maintained systems, resolved hardware/software/network issues, ensured operational continuity.

Education

B.Sc. Information Technology (Expected Sept 2026)
University of Embu, Kenya

- **Relevant Coursework:** AI, Software Engineering, Web Development, Mobile Development, Data Structures, Cloud Computing
- **Honors:** Innovation Leadership Recognition (2024)

Leadership & Community

- **GDG on Campus Organizer** (2022–Present)
  Co-organized THiNK Botathon, ICP Blockchain, Moringa Cybersecurity Bootcamp, Agentic AI Workshop, Droidcon Ke26.

Languages

- **English:** Full professional proficiency | **Swahili:** Fluent